Simulation study

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This document aims to reproduce the simulation study 'Introduction to statistical simulations in health research' by (Boulesteix et al. 2020).

```
#load packages
#| echo: false
#| output: false
suppressMessages({
  suppressWarnings({
    library(Hmisc)
    library(mice)
    library(tidyverse)
  })
})
```

Data prep

The five data sets are being loaded using the following code:

```
d1 <- sasxport.get("../data/DEMO_I.xpt")
d2 <- sasxport.get("../data/BPX_I.xpt")
d3 <- sasxport.get("../data/BMX_I.xpt")
d4 <- sasxport.get("../data/GHB_I.xpt")
d5 <- sasxport.get("../data/TCHOL_I.xpt")
d1.t <- subset(d1,select=c("seqn","riagendr","ridageyr"))
d2.t <- subset(d2,select=c("seqn","bpxsy1"))
d3.t <- subset(d3,select=c("seqn","bmxbmi"))
d4.t <- subset(d4,select=c("seqn","lbxgh"))
d5.t <- subset(d5,select=c("seqn","lbdtcsi"))
d1.t <- subset(d1,select=c("seqn","riagendr","ridageyr"))
```

```r
d2.t <- subset(d2,select=c("seqn","bpxsy1"))
d3.t <- subset(d3,select=c("seqn","bmxbmi"))
d4.t <- subset(d4,select=c("seqn","lbxgh"))
d5.t <- subset(d5,select=c("seqn","lbdtcsi"))
d <- merge(d1.t,d2.t)
d <- merge(d,d3.t)
d <- merge(d,d4.t)
d <- merge(d,d5.t)
```

Renaming the variables to be more comprehensive:

```r
d$age <- d$ridageyr
d$sex <- d$riagendr
d$bp <- d$bpxsy1
d$bmi <- d$bmxbmi
d$HbA1C <- d$lbxgh
d$chol <- d$lbdtcsi
```

Minors are set to NA and only complete cases are being kept:

```r
d$age[d$age<18] <- NA
dc <- cc(subset(d,select=c("age","sex","bmi","HbA1C","bp")))
```

Analysis

Assessing the impact of Glycohemoglobin level on systolic blood pressure assuming no measurement errors

A linear regression is run with systolic blood pressure *bp* as the dependent variable and Glycohemoglobin level *HbA1C* as the main predictor, adjusted for age and gender.

```r
summary(lm(bp ~ HbA1C + age + as.factor(sex), data=dc))
```

```
Call:
lm(formula = bp ~ HbA1C + age + as.factor(sex), data = dc)

Residuals:
Systolic: Blood pres (1st rdg) mm Hg
    Min      1Q  Median      3Q     Max
```

```
-49.887 -10.509  -1.378   8.491 107.583

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     98.75149    1.21418  81.332  < 2e-16 ***
HbA1C            1.12638    0.20291   5.551 2.98e-08 ***
age              0.44486    0.01284  34.648  < 2e-16 ***
as.factor(sex)2 -3.24792    0.45164  -7.191 7.34e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16.1 on 5088 degrees of freedom
Multiple R-squared:  0.2305,    Adjusted R-squared:   0.23
F-statistic:   508 on 3 and 5088 DF,  p-value: < 2.2e-16
```

TThe coefficient for Glycohemoglobin (HbA1C) on systolic blood pressure, controlling for age and sex, is extracted:

```r
round(lm(bp ~ HbA1C + age + as.factor(sex), data=dc)$coef[2],2)
```

```
HbA1C
 1.13
```

The 95% confidence interval of the regression parameters:

```r
confint(lm(bp ~ HbA1C + age + as.factor(sex), data=dc))
```

```
                     2.5 %      97.5 %
(Intercept)     96.3711755 101.1317982
HbA1C            0.7285836   1.5241825
age              0.4196932   0.4700355
as.factor(sex)2 -4.1333281  -2.3625106
```

The second regression with the Body Mass Index of the respondent *bmi* as a confounder.:

```r
summary(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc))
```

```
Call:
```

```
lm(formula = bp ~ HbA1C + bmi + age + as.factor(sex), data = dc)

Residuals:
Systolic: Blood pres (1st rdg) mm Hg
    Min      1Q  Median      3Q     Max
-51.068 -10.251  -1.504   8.264 107.410

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     92.65583    1.39320  66.506  < 2e-16 ***
HbA1C            0.75177    0.20596   3.650 0.000265 ***
bmi              0.28632    0.03282   8.724  < 2e-16 ***
age              0.44586    0.01275  34.979  < 2e-16 ***
as.factor(sex)2 -3.63115    0.45049  -8.060  9.4e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.98 on 5087 degrees of freedom
Multiple R-squared:  0.2418,    Adjusted R-squared:  0.2412
F-statistic: 405.7 on 4 and 5087 DF,  p-value: < 2.2e-16
```

The effect of Glycohemoglobin on systolic blood pressure, adjusted for age, sex, and BMI, is reported as:

```r
round(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc)$coef[2],2)
```

```
HbA1C
 0.75
```

With the CI of:

```r
confint(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc))
```

```
                     2.5 %     97.5 %
(Intercept)     89.9245592 95.3871089
HbA1C            0.3479966  1.1555348
bmi              0.2219815  0.3506673
age              0.4208695  0.4708464
as.factor(sex)2 -4.5143014 -2.7479929
```

Assessing the impact of Glycohemoglobin on systolic blood pressure in the presence of measurement errors

The goal is to assess how measurement errors in both the exposure and the confounder affect the regression results.

The reference value for the impact of Glycohemoglobin on systolic blood pressure, without measurement error:

```
ref <- lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc)$coef[2]
```

HbA1C.me and *bmi.me* are then created in a simulation process of 1000 datasets with variance of the measurement errors between 0 and 0.5 of their original variances:

```
n.sim <- 1e3
perc.me.exp <- seq(0,.5,.1)
perc.me.conf<- seq(0,.5,.1)
scenarios <- expand.grid(perc.me.exp,perc.me.conf)
```

The reference variances in the original exposure and confounder variables are stored in "var.exp" and "var.conf":

```
var.exp <- var(dc$HbA1C)
var.conf <- var(dc$bmi)
```

The original sample size:

```
n <- dim(dc)[1]
```

The empty matrix "beta.hat" is created to store the regression coefficients of *HbA1C.me*.

```
beta.hat <- matrix(ncol=dim(scenarios)[1], nrow=n.sim)
```

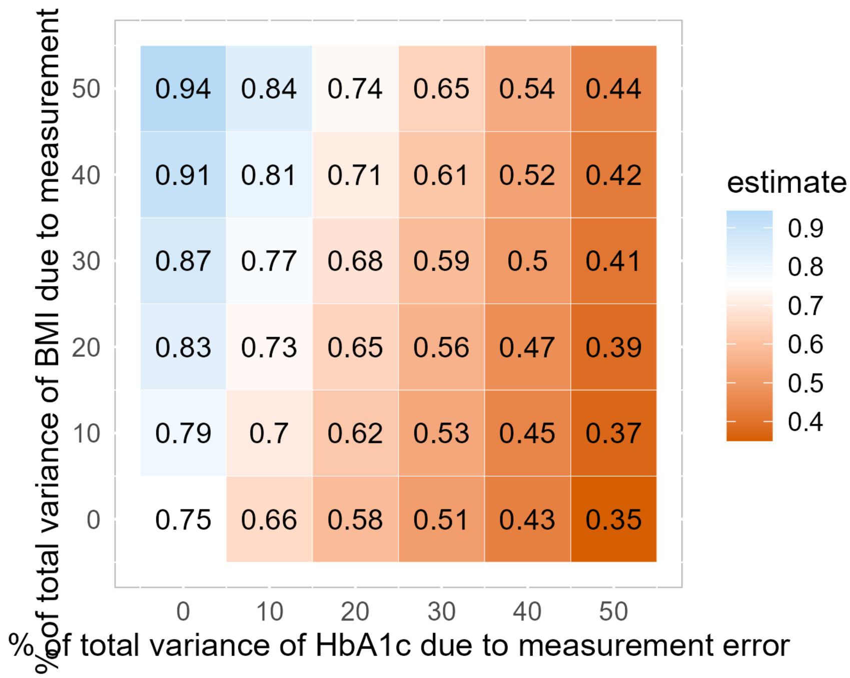
The simulation process involves nested loops for each of the 1000 simulations and 36 scenarios, where *HbA1C.me* and *bmi.me* are generated by adding measurement errors randomly sampled from normal distributions:

```
for (k in 1:n.sim){
  print(k)
  set.seed(k)
  for (i in 1:dim(scenarios)[1]){
    var.me.exp <- var.exp*scenarios[i,1]/(1-scenarios[i,1])
    var.me.conf <- var.conf*scenarios[i,2]/(1-scenarios[i,2])
```

```
    dc$HbA1C.me <- dc$HbA1C + rnorm(dim(dc)[1], 0, sqrt(var.me.exp) )
    dc$bmi.me <- dc$bmi + rnorm(dim(dc)[1], 0, sqrt(var.me.conf) )
    beta.hat[k,i] <- lm(bp ~ HbA1C.me + age + bmi.me + as.factor(sex), data=dc)$coef[2]
  }}
```

Results

The result of the simulation is visualized using a heatmap showing how the impact of Glycohemoglobin on systolic blood pressure varies according to different levels of residual variance for both the exposure *HbA1C* and confounder *BMI*. The data is stored in the tot.mat matrix and plotted using the ggplot2 package:



The figure matches the figure obtained by Boulesteix, Groenwold, Abrahamowicz, et al. (2020).

References

Boulesteix, Anne-Laure, Rolf Hh Groenwold, Michal Abrahamowicz, Harald Binder, Matthias Briel, Roman Hornung, Tim P Morris, Jörg Rahnenführer, and Willi Sauerbrei. 2020. "Introduction to Statistical Simulations in Health Research." *BMJ Open* 10 (12): e039921. https://doi.org/10.1136/bmjopen-2020-039921.